

Class18_MiniProject

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###Pertussis and the CMI-PB project

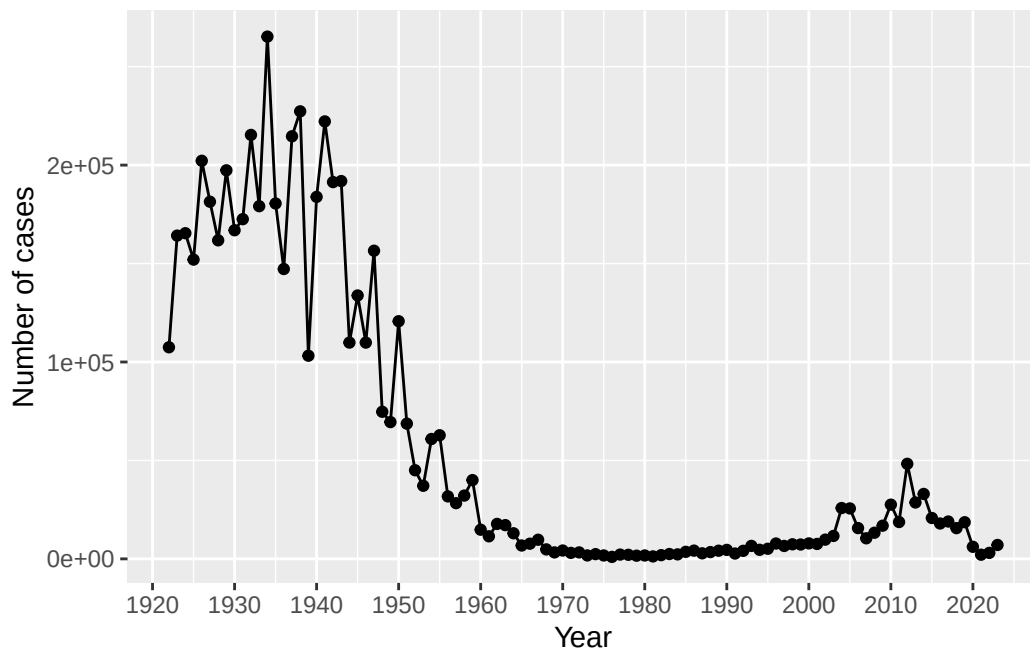
##Investigating pertussis cases by year

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
cdc <- read.csv("cdc.csv")
cdc$No..Reported.Pertussis.Cases <- as.numeric(gsub(",", "", cdc$No..Reported.Pertussis.Cases))

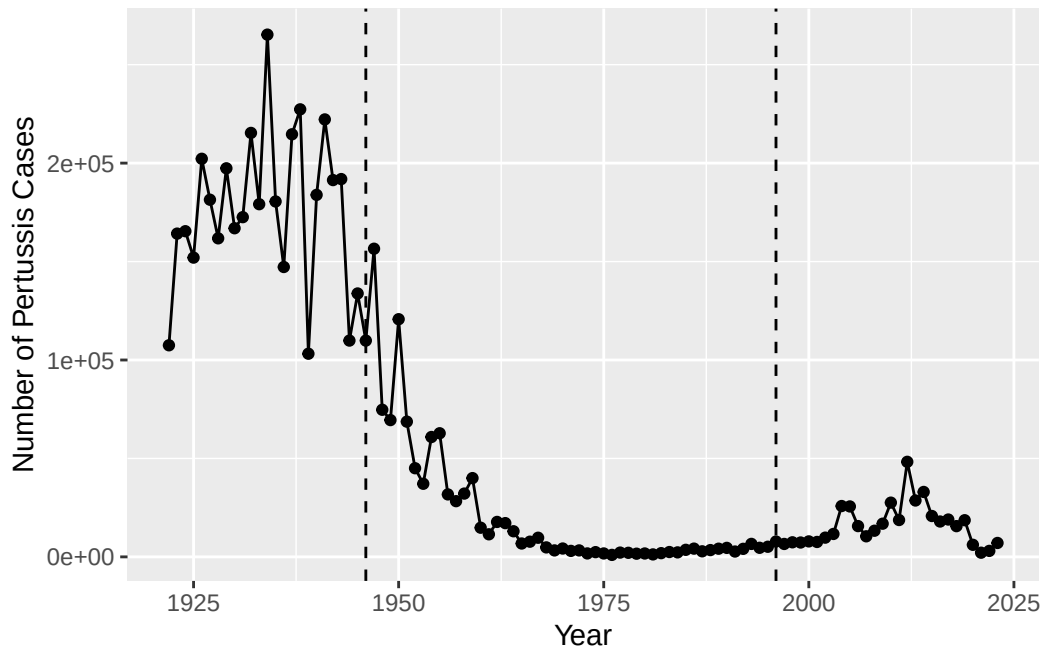
library(ggplot2)

ggplot(cdc) +
  aes(x = Year, y = No..Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  scale_x_continuous(breaks = seq(1920, 2020, by = 10)) +
  labs(x = "Year", y = "Number of cases")
```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(x = Year, y = No..Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, linetype = "dashed") +
  geom_vline(xintercept = 1996, linetype = "dashed") +
  labs(
    x = "Year",
    y = "Number of Pertussis Cases"
  )
)
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the acellular pertussis (aP) vaccine was introduced in 1996, reported pertussis cases began to increase again after many years of very low incidence. The number of cases gradually rose through the 2000s, with a large spike in 2012, when about 48,277 cases were reported—the highest number since 1955. Several explanations have been proposed for this resurgence. One possibility is that PCR-based diagnostic tests became more sensitive, allowing more infections to be detected and reported. Another factor may be vaccine hesitancy, which reduces vaccination coverage in some communities. In addition, bacterial evolution may allow the pathogen to partially escape immunity generated by vaccines. Finally, immunity from the aP vaccine may wane more quickly than the older whole-cell (wP) vaccine, especially in adolescents who were vaccinated as infants. These factors together may contribute to the observed increase in pertussis cases

##The CMI-PB API returns JSON data

```
# Allows us to read, write and process JSON data
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

```
subject_id infancy_vac biological_sex ethnicity race
```

1	1	wP	Female Not Hispanic or Latino White
2	2	wP	Female Not Hispanic or Latino White
3	3	wP	Female Unknown White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female  Male
112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

	American Indian/Alaska Native	Asian	Black or African American
Female	0	32	2
Male	1	12	3

	More Than One Race	Native Hawaiian or Other Pacific Islander
Female	15	1
Male	4	1

	Unknown or Not Reported	White
Female	14	48
Male	7	32

Working with data

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
today()
```

```
[1] "2026-03-11"
```

```
today() - ymd("2000-01-01")
```

Time difference of 9566 days

```
time_length( today() - ymd("2000-01-01"), "years")
```

```
[1] 26.19028
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(lubridate)
```

```
subject$age <- today() - ymd(subject$year_of_birth)
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
ap <- subject %>% filter(infancy_vac == "aP")
round(summary(time_length(ap$age, "years")))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>% filter(infancy_vac == "wP")
round(summary(time_length(wp$age, "years")))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

Average age of aP individuals: ~ 26 years

Average age of wP individuals: ~ 36 years

Q8. Determine the age of all individuals at time of boost?

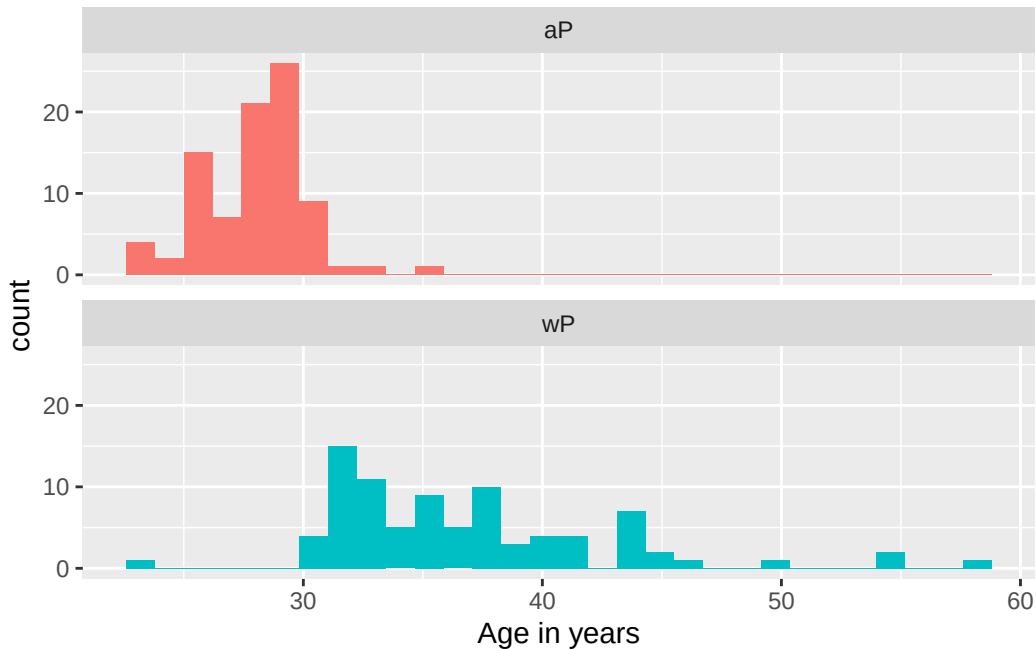
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
       fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
# Or use wilcox.test()
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

##Joining multiple tables

```
library(jsonlite)
library(dplyr)

# Read specimen table
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)

# Read titer/ab_titer table
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
library(dplyr)

# Q9: join specimen and subject
meta <- left_join(specimen, subject, by = "subject_id")

# check dimensions and first few rows
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                      -3
2           2           1                       1
3           3           1                       3
4           4           1                       7
5           5           1                      11
6           6           1                      32
planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                0          Blood      1         wP         Female
2                1          Blood      2         wP         Female
3                3          Blood      3         wP         Female
4                7          Blood      4         wP         Female
5               14          Blood      5         wP         Female
6               30          Blood      6         wP         Female
ethnicity race year_of_birth date_of_boost dataset
1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
age
1 14679 days
2 14679 days
3 14679 days
4 14679 days
5 14679 days
6 14679 days
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
library(dplyr)

# Q10: join titer with meta data
abdata <- inner_join(titer, meta, by = "specimen_id")

# check dimensions
dim(abdata)
```

```
[1] 52576    21
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
 IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  5389 10117 10124 10124 10124
```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
# List all unique dataset values
unique(abdata$dataset)
```

```
[1] "2020_dataset" "2021_dataset" "2022_dataset" "2023_dataset"
```

```
# Count number of rows per dataset
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
          31520          8085          7301          5670
```

##Examine IgG Ab titer levels

Now using our joined/merged/linked abdata dataset filter() for IgG isotype.

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

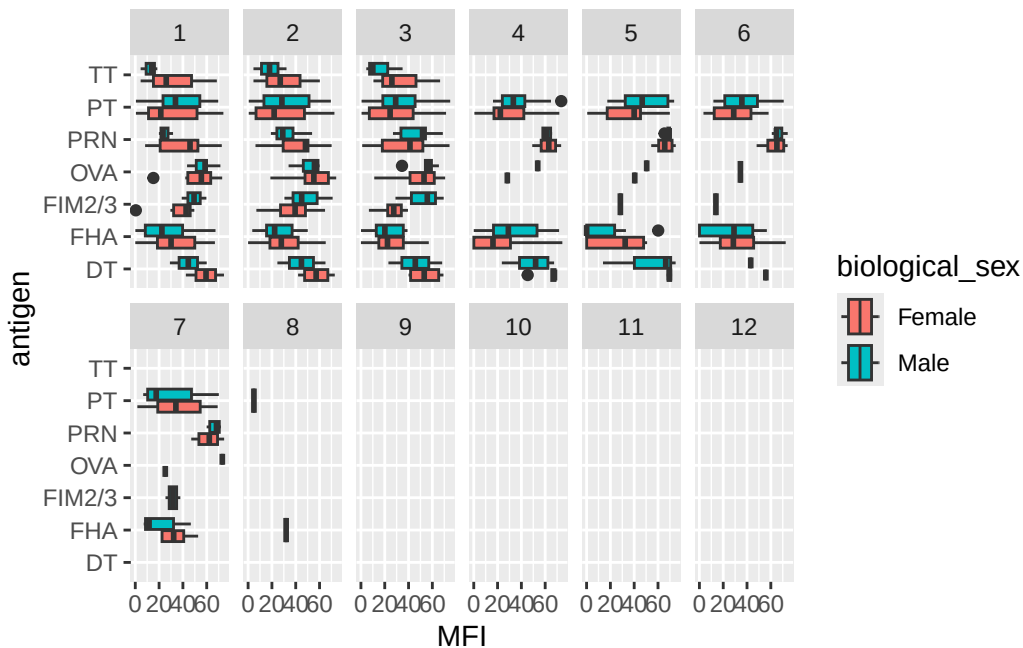
	age
1	14679 days
2	14679 days
3	14679 days
4	15775 days
5	15775 days
6	15775 days

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
library(ggplot2)

ggplot(igg) +
  aes(x = MFI, y = antigen, fill = biological_sex) +
  geom_boxplot() +
  xlim(0, 75) +
  facet_wrap(vars(visit), nrow = 2)
```

Warning: Removed 4475 rows containing non-finite outside the scale range (`stat\_boxplot()`).

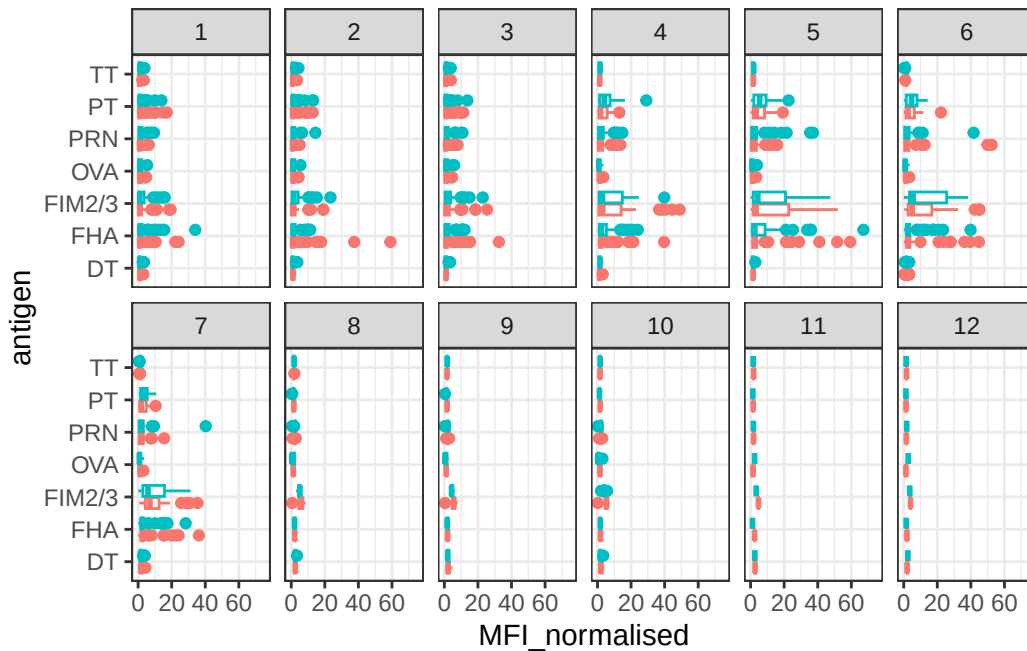


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others? IgG titers change over time primarily for vaccine-targeted antigens like PT, PRN, and FHA, because these are immunodominant and included in the vaccine, whereas other antigens show little change.

We can attempt to examine differences between wP and aP here by setting color and/or facet values of the plot to include infancy_vac status (see below). However these plots tend to be rather busy and thus hard to interpret easily.

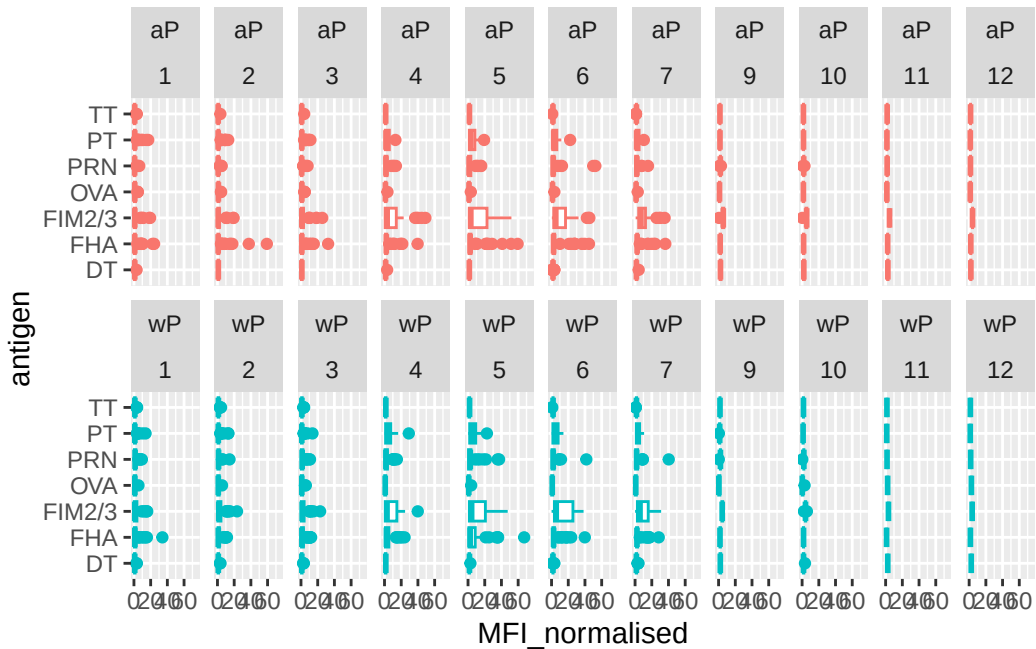
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

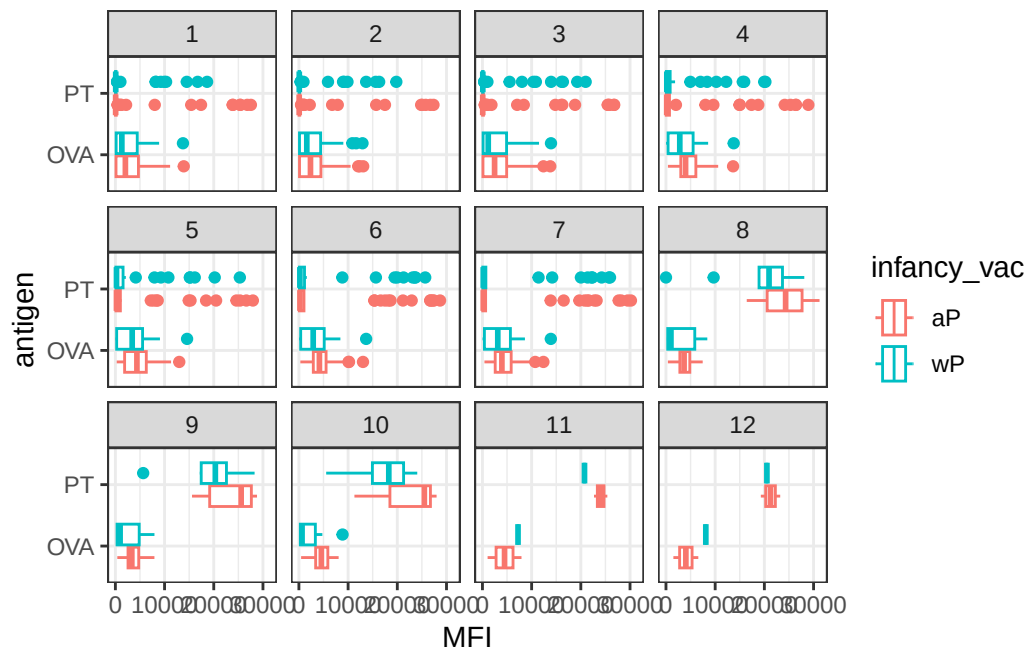
Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each.

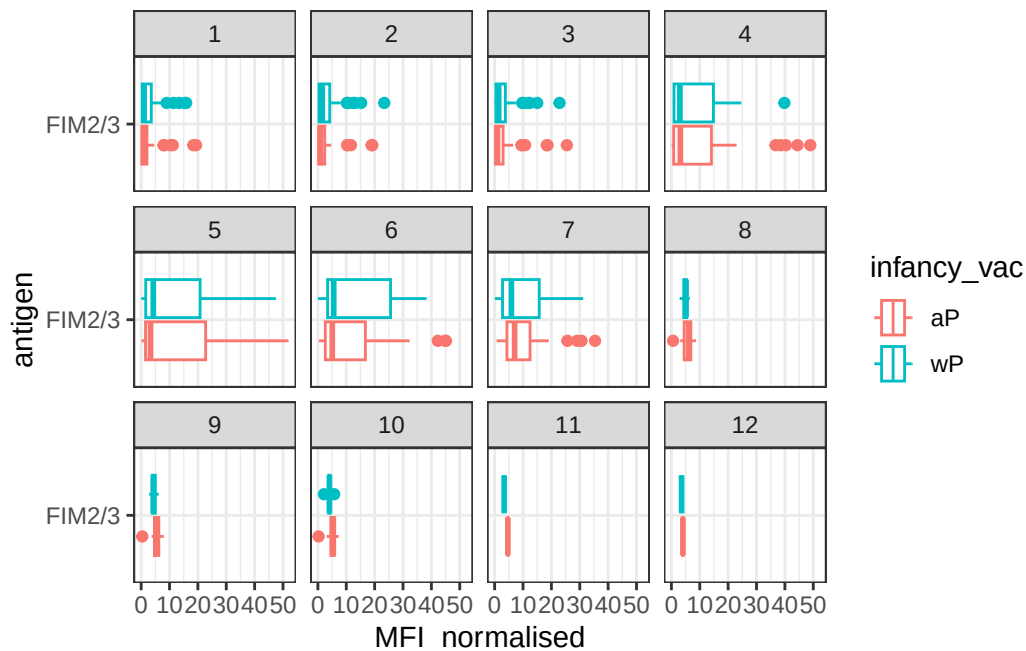
```
library(dplyr)
library(ggplot2)

# Filter for OVA and PT
igg %>%
  filter(antigen %in% c("OVA", "PT")) %>%
  ggplot() +
  aes(x = MFI, y = antigen, col = infancy_vac) + # MFI on x-axis, antigen on y-axis, color by visit
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
library(dplyr)
library(ggplot2)

igg %>%
  filter(antigen == "FIM2/3") %>%
  ggplot() +
  aes(x = MFI_normalised, y = antigen, col = infancy_vac) + # normalized MFI on x, antigen on y
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

PT shows a vaccine-driven, time-dependent increase in IgG, whereas the control antigen OVA shows no change, highlighting the specificity and effectiveness of the immune response.

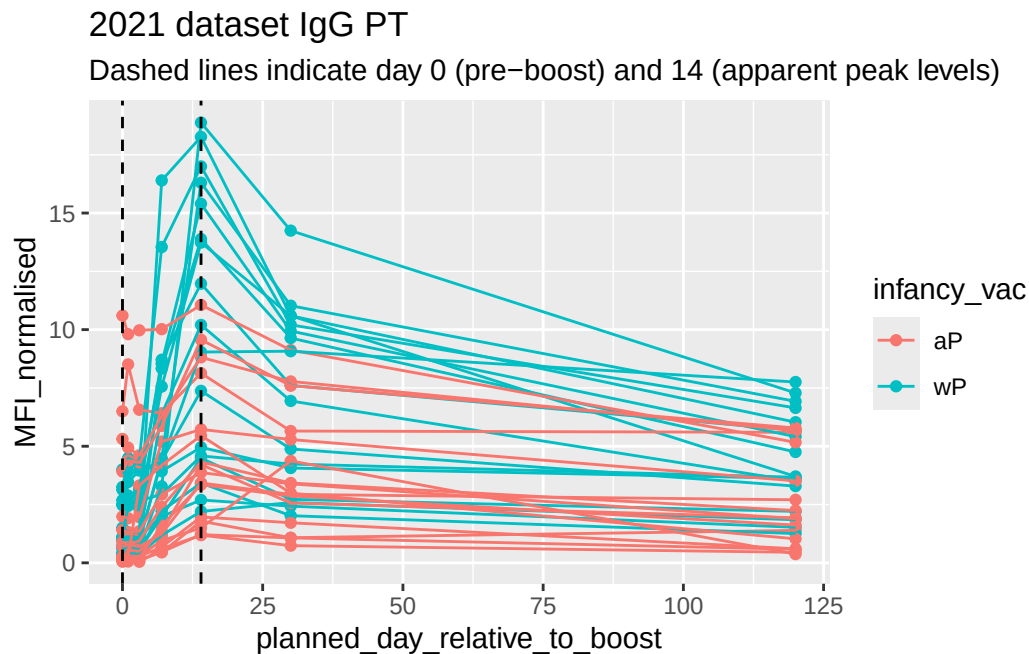
Q17. Do you see any clear difference in aP vs. wP responses?

wP vaccination tends to produce stronger, more sustained IgG responses to PT, whereas aP vaccination leads to lower or more variable responses.

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
```

```
labs(title="2021 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



Q18. Does this trend look similar for the 2020 dataset?

The overall trend is very similar between 2020 and 2021 datasets

##Obtaining CMI-PB RNASeq data

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)

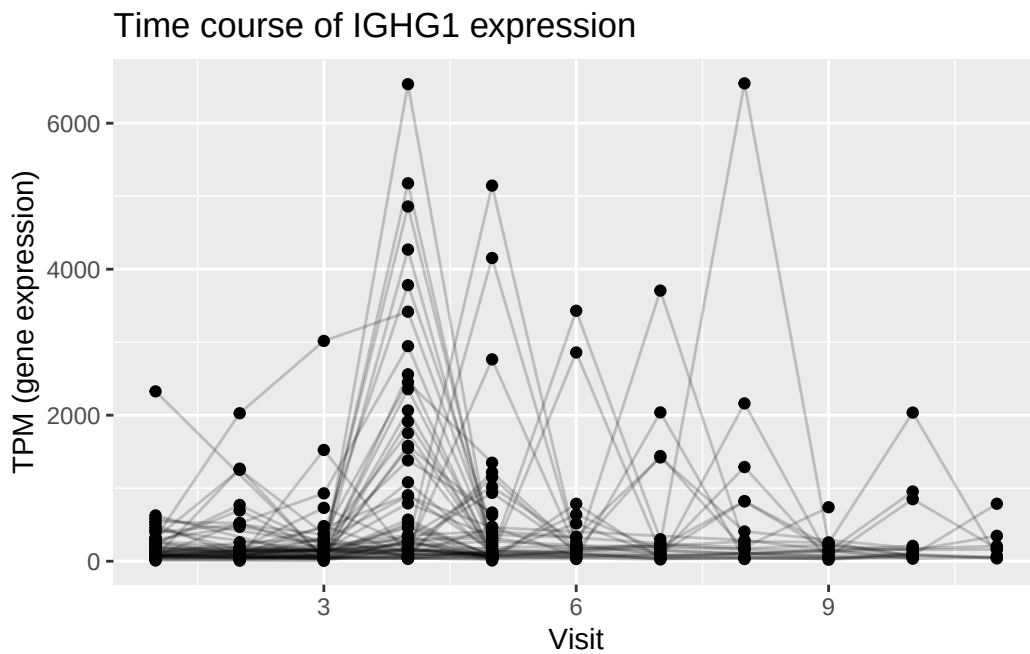
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
library(ggplot2)

ggplot(ssrna) +
  aes(x = visit, y = tpm, group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.2) +
  labs(title = "Time course of IGHG1 expression",
       x = "Visit",
       y = "TPM (gene expression)")
```



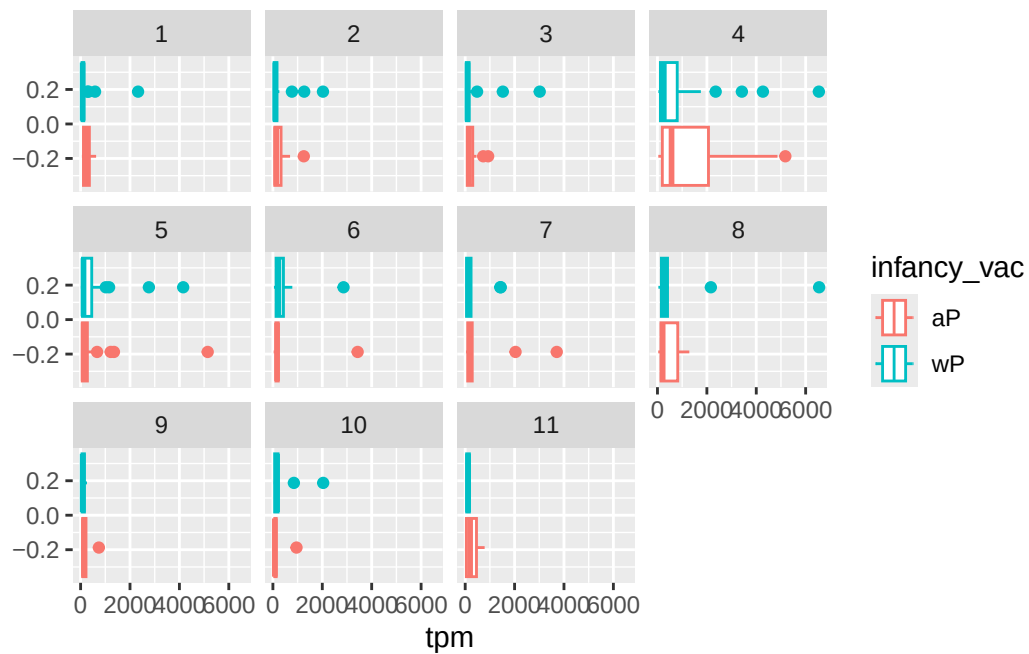
Q20. What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

Maximum IGHG1 expression usually occurs earlier than the peak of circulating IgG1 antibody, reflecting that mRNA expression precedes protein production.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

Transcription is upstream of protein production; RNA levels rise rapidly post-stimulation, while circulating antibody levels are delayed due to protein synthesis, secretion, and half-life in plasma.

```
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```



```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
  aes(tpm, col=infancy_vac) + geom_density() +
  geom_rug()
```

